

DP Computer Science: A 3.4.3 OLAP and Data Mining (HL Only)

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Explain the role of **Online Analytical Processing (OLAP)** in business intelligence
- Describe the function of **data mining** in extracting meaningful information from large data sets
- Identify and explain various **data mining techniques**, including classification, clustering, regression, association rule discovery, sequential pattern discovery, and anomaly detection
- Discuss the uses of OLAP and different data mining techniques in **real-world business intelligence** scenarios

Key Vocabulary

Term	Definition
OLAP	Online Analytical Processing — technology that allows users to analyse multidimensional data interactively
Data mining	The process of discovering patterns and extracting meaningful information from large datasets
Business Intelligence (BI)	The use of data analysis to support business decision-making
Classification	A data mining technique that categorises data into predefined classes (e.g., spam/not spam)
Clustering	A data mining technique that groups similar data points together without predefined categories
Regression	A data mining technique that predicts continuous numeric values (e.g., sales forecasting)
Association rules	Finding relationships between variables (e.g., "customers who buy X also buy Y")
Sequential patterns	Discovering patterns in sequences of events over time (e.g., purchase sequences)
Anomaly detection	Identifying unusual or outlier data points (e.g., fraud detection)
Machine learning	Algorithms that allow systems to learn from data and make predictions
Supervised learning	Machine learning using labelled data
Unsupervised learning	Machine learning using unlabelled data

Relevant ATL Skills

Skill	Description
Thinking	Critical thinking (analysing value and limitations); information literacy; problem-solving
Communication	Using appropriate terminology; discussing and presenting examples
Research	Gathering information on data mining techniques; investigating real-world applications

2. The Data Deluge and Business Intelligence

Why Do We Need OLAP and Data Mining?



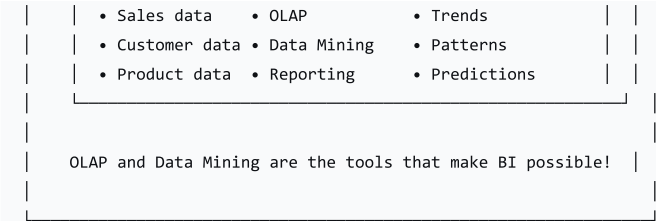
What is Business Intelligence (BI)?

Business Intelligence (BI) is the use of data analysis to support business decision-making. It involves collecting, analysing, and presenting data to help organisations make informed decisions.

Goal: Transform raw data into actionable insights that drive business value.

How BI Works





3. Online Analytical Processing (OLAP)

What is OLAP?

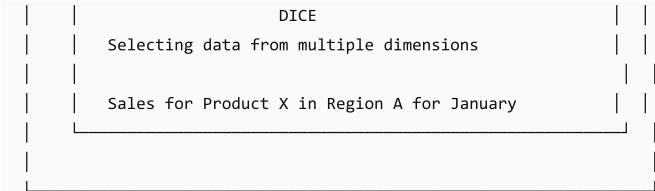
OLAP (Online Analytical Processing) is technology that allows users to analyse multidimensional data interactively. It enables rapid answers to complex analytical queries.

Key Features of OLAP

Feature	Description
Multidimensional	Data is organised in multiple dimensions (time, product, region)
Interactive	Users can explore data through drill-down, roll-up, slice, and dice
Fast Query Response	Pre-calculated summaries provide quick answers
Historical Analysis	Supports trend analysis and forecasting
Aggregated Data	Data is summarised for high-level analysis

OLAP Operations





Real-World OLAP Example

Example	Explanation
Sales Performance Analysis	Total sales by region, by product, by month
Year-over-Year Comparison	How did this quarter compare to last year?
Product Line Performance	Which products are performing best?
Regional Performance	Which regions need more support?

4. Data Mining

What is Data Mining?

Data mining is the process of discovering patterns and extracting meaningful information from large datasets. It involves using statistical and machine learning techniques to find hidden insights.

The Data Mining Process



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Step 5: Deployment
Using the insights to make business decisions

5. Data Mining Techniques

Overview of Techniques

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DATA MINING TECHNIQUES

CLASSIFICATION

Categories data into predefined classes
Example: Spam detection, credit risk assessment

CLUSTERING

Groups similar data points without predefined labels
Example: Customer segmentation, market research

REGRESSION

Predicts continuous numeric values
Example: Sales forecasting, price prediction

ASSOCIATION RULES

Finds relationships between variables
Example: Market basket analysis (beer and diapers)

SEQUENTIAL PATTERNS

Discovers patterns in sequences of events
Example: Customer purchase sequences, website navigation patterns

ANOMALY DETECTION

Identifies unusual or outlier data points
Example: Fraud detection, system failure prediction

5.1 Classification

| Classification categorises data into predefined classes or categories.

Aspect	Description
Type	Supervised learning
Input	Labelled training data
Output	Category/class label
Example	"Is this email spam or not spam?"
Application	Description
Spam Filtering	Classify emails as spam or legitimate
Credit Scoring	Classify loan applicants as high/low risk
Medical Diagnosis	Classify tumours as benign or malignant
Image Recognition	Classify images as cat, dog, etc.

Analogy: Classification is like sorting laundry into categories (whites, colours, delicates).

5.2 Clustering

| Clustering groups similar data points together without using predefined categories.

Aspect	Description
Type	Unsupervised learning
Input	Unlabelled data
Output	Groups/clusters
Example	"Which customers have similar shopping habits?"
Application	Description
Customer Segmentation	Group customers by behaviour, preferences, demographics
Market Research	Identify distinct market segments
Social Network Analysis	Find communities within social networks
Document Classification	Group similar documents together

Analogy: Clustering is like organising a library by topic when the books don't have labels—you find what belongs together naturally.

5.3 Regression

| Regression predicts continuous numeric values based on input data.

Aspect	Description
Type	Supervised learning
Input	Labelled training data
Output	Continuous numeric value

Aspect	Description
Example	"What will next month's sales be?"
Application	Description
Sales Forecasting	Predict future sales based on past trends
Price Prediction	Estimate house prices, stock prices
Demand Forecasting	Predict product demand
Risk Assessment	Predict probability of default

Analogy: Regression is like predicting the final exam score from a student's performance throughout the term.

5.4 Association Rules

Association rules find relationships or correlations between variables in a dataset.

Aspect	Description
Type	Unsupervised learning
Input	Transaction data
Output	Rules ($X \rightarrow Y$)
Example	"Customers who buy bread also buy milk"
Application	Description
Market Basket Analysis	Identify products frequently bought together
Cross-selling	Recommend complementary products
Inventory Management	Understand which products are often purchased together
Website Recommendations	Recommend content based on user behaviour

Analogy: Association rules are like noticing that people who buy chips often also buy dip—they go together.

5.5 Sequential Patterns

Sequential patterns discover patterns in sequences of events over time.

Aspect	Description
Type	Unsupervised learning
Input	Sequence data with time stamps
Output	Sequences of events
Example	"Customers who buy a smartphone often buy a case within a week"
Application	Description
Purchase Sequences	Understand order of product purchases

Application	Description
Web Navigation	Analyse user browsing paths
Customer Journey	Map customer interactions over time
Predictive Analysis	Anticipate what a user might do next

Analogy: Sequential patterns are like following a recipe—each step leads to the next.

5.6 Anomaly Detection

 Anomaly detection identifies unusual or outlier data points that do not fit expected patterns.

Aspect	Description
Type	Unsupervised learning (often)
Input	Data with expected patterns
Output	Anomalous data points
Example	"Is this transaction suspicious?"
Application	Description
Fraud Detection	Identify unusual credit card transactions
Network Security	Detect unusual network traffic (intrusions)
Quality Control	Identify defective products
System Monitoring	Detect server failures before they happen

Analogy: Anomaly detection is like a security guard noticing someone who doesn't look like they belong—something is different.

6. Classification vs. Clustering vs. Regression

Feature	Classification	Clustering	Regression
Type	Supervised	Unsupervised	Supervised
Outcome	Category/Class	Group/Cluster	Numeric Value
Example	Spam/Not Spam	Customer Segments	Sales Prediction
Labels Needed	Yes	No	Yes
Goal	Assign to predefined class	Find natural groupings	Predict numeric value

7. OLAP vs. Data Mining

Feature	OLAP	Data Mining
Purpose	Interactive analysis of data	Discovering hidden patterns

Feature	OLAP	Data Mining
Approach	Exploratory (user-driven)	Automatic (algorithm-driven)
Output	Summarised views, charts	Models, rules, patterns
Questions	"What happened?"	"Why did it happen? What will happen?"
Methods	Drill-down, roll-up, slice, dice	Classification, clustering, association
Example	Monthly sales report	Predicting future sales trends

The Big Picture

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OLAP AND DATA MINING TOGETHER

OLAP answers: "What happened?"
↳ e.g., "Our sales increased by 15% in Q3"

Data Mining answers: "Why did it happen?"
↳ e.g., "The increase was driven by a new marketing campaign targeting customers aged 25-34 who had previously browsed but not purchased."

Together, they provide a complete BI solution!

8. Real-World Business Intelligence Examples

Business Scenario	OLAP Use	Data Mining Use
Retail	Monthly sales by product category	Association rules for product recommendations
Banking	Transaction volume by region	Fraud detection (anomaly detection)
E-commerce	Customer order history	Customer segmentation (clustering)
Healthcare	Patient outcomes by treatment	Predictive analysis (regression)
Manufacturing	Production output by machine	Quality control (anomaly detection)
Marketing	Campaign performance by channel	Customer response prediction (classification)

9. Ethical Considerations

Consideration	Description
Privacy	Data mining often uses personal data; privacy must be respected
Consent	Users should know how their data is being used
Bias	Algorithms can perpetuate or amplify existing biases
Transparency	Users should understand how decisions are made
Data Security	Collected data must be protected from breaches
Accountability	Who is responsible for data mining outcomes?

10. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Introduction	5 min	Data deluge; the need for insights
Business Intelligence	5 min	Definition and goal of BI
OLAP	10 min	Definition, key features, operations (roll-up, drill-down, slice, dice)
Data Mining	5 min	Definition, goal, link to Machine Learning
Data Mining Techniques	15 min	Classification, Clustering, Regression, Association Rules, Sequential Patterns, Anomaly Detection
BI Uses	5 min	Real-world applications
Ethical Considerations	5 min	Privacy, bias, transparency
Worksheet/Discussion	5 min	Applying concepts to scenarios
Conclusion	5 min	Recap; Q&A

Discussion Questions

Question	Purpose
"How could a supermarket use association rules?"	Application of association rules
"What is the difference between classification and clustering?"	Distinguish supervised vs. unsupervised
"When would you use OLAP instead of data mining?"	Understand different uses
"What are some ethical concerns with data mining?"	Discuss ethical implications

11. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Worksheet	Evaluate understanding of OLAP and data mining techniques
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the difference between OLAP and data mining?
2. What is the goal of classification?
3. Give an example of association rules.
4. Why is anomaly detection important in fraud detection?

12. Differentiation

Level	Strategy
Support	Provide definitions and matching exercises; use relatable real-world examples
Challenge	Research specific case studies; explore implementation details

13. Resources

- **Presentation** (Slide Deck)
- **eBook** (A3.4.3 with examples)
- **Worksheet:** OLAP and data mining scenarios
- **Worksheet Answers**

14. Summary: Key Takeaways

Concept	Key Point
OLAP	Interactive multidimensional analysis; answers "what happened?"
Data Mining	Automatic pattern discovery; answers "why?" and "what will happen?"
Classification	Predicts categories (spam, credit risk)
Clustering	Groups similar data (customer segments)
Regression	Predicts numeric values (sales forecasting)
Association Rules	Finds relationships (market basket analysis)
Sequential Patterns	Finds event sequences (customer journeys)
Anomaly Detection	Finds outliers (fraud detection)

KEY REMINDERS

- ✓ OLAP = Interactive analysis of data
- ✓ Data Mining = Automatic pattern discovery
- ✓ Classification = Predicts categories
- ✓ Clustering = Groups similar data
- ✓ Regression = Predicts numbers
- ✓ Association Rules = Finds relationships
- ✓ Sequential Patterns = Discovers event sequences
- ✓ Anomaly Detection = Finds outliers

"OLAP and data mining are essential tools for turning raw data into actionable business intelligence—helping organisations understand the past, see the present, and predict the future."